Recommandations et Standards de Pratique du CIRSE sur l'ablation thermique des tumeurs osseuses

Titre original : CIRSE Standards of Practice on Thermal Ablation of Bone Tumours

Auteurs : Ryan A, Byrne C et al. | **Revue :** *Cardiovascular and interventional radiology* (**IF : 2.8**) | **Date :** Mai 2022 | **PMID :** 35348870 | **Modalité :** Scanner TDM / IRM / Fluoroscopie

1. Pertinence Clinique & Lien avec l'Agenda Interventionnel (Cochin)

Issu de votre collection Zotero « Ablation percutanée » (Ryan, Tsoumakidou, Buy, Gangi et al., *CVIR*, mai 2022) — Référentiel européen officiel encadrant les standards techniques, la thermoprotection et les indications d'ablation osseuse à Cochin.

2. Synthèse Clinique en Français

Résumé exécutif : Consensus officiel du CIRSE (*CardioVascular and Interventional Radiology*, mai 2022) définissant les standards de bonne pratique pour l'ablation thermique percutanée des tumeurs osseuses bénignes et malignes : indications curatives et palliatives, choix de l'énergie, monitorage thermique et gestes adjuvants (embolisation, consolidation).

Analyse détaillée : Rédigé par le comité des Standards de Pratique du CIRSE, ce document de référence établit les recommandations européennes pour la réalisation sécurisée des ablations thermiques osseuses (radiofréquence, cryoablation, micro-ondes, laser). Il précise le bilan pré-thérapeutique, les seuils de sécurité thermique pour la moelle épinière et les nerfs périphériques, les techniques de protection active (hydrodissection, carbodissection, thermocouples, potentiels évoqués moteurs/sensitifs) et le timing optimal de l'embolisation pré-opératoire (tumeurs hypervasculaires : rein, thyroïde) et de la cimentoplastie.

3. Protocole Technique & Paramètres Opératoires

Validation préalable en RCP • Bilan d'imagerie en coupes (< 1 mois) • Antibioprophylaxie ciblée (céphalosporine 1re/2e génération, surtout si cimentoplastie associée) • Thermométrie par thermocouple + dissection isolante dès qu'une structure nerveuse est à < 10 mm • Consolidation PMMA immédiate en zone portante.

4. Points Clés à Retenir pour la Pratique

- • Référentiel européen officiel du CIRSE pour les ablations thermiques osseuses bénignes (ostéome ostéoïde, chondroblastome) et malignes (métastases).
- • Recommandations précises sur le choix entre RF, Cryoablation et Micro-ondes selon le caractère ostéolytique ou ostéocondensant et la taille tumorale.
- • Standardisation de la thermoprotection active (CO2, sérum glucosé/salé, contrôle par thermocouple avec seuils stricts < 10 °C et > 45 °C).
- • Indication d'embolisation transartérielle préalable pour les volumineuses métastases hypervasculaires afin de réduire le saignement et l'effet 'heat-sink'.

5. Original PubMed Abstract (Texte Intégral de l'Abstract)

Background Percutaneous thermal ablation is an effective, minimally invasive means of treating a variety of focal benign and malignant osseous lesions. To determine the role of ablation in individual cases, multidisciplinary team (MDT) discussion is required to assess the suitability and feasibility of a thermal ablative approach, to select the most appropriate technique and to set the goals of treatment i.e. curative or palliative. Purpose This document will presume the indication for treatment is clear and approved by the MDT and will

define the standards required for the performance of each modality. CIRSE Standards of Practice documents are not intended to impose a standard of clinical patient care, but recommend a reasonable approach to, and best practices for, the performance of thermal ablation of bone tumours. Methods The writing group was established by the CIRSE Standards of Practice Committee and consisted of five clinicians with internationally recognised expertise in thermal ablation of bone tumours. The writing group reviewed the existing literature on thermal ablation of bone tumours, performing a pragmatic evidence search using PubMed to search for publications in English and relating to human subjects from 2009 to 2019. Selected studies published in 2020 and 2021 during the course of writing these standards were subsequently included. The final recommendations were formulated through consensus. Results Recommendations were produced for the performance of thermal ablation of bone tumours taking into account the biologic behaviour of the tumour and the therapeutic intent of the procedure. Recommendations are provided based on lesion characteristics and thermal modality, for the use of tissue monitoring and protection, and for the appropriately timed application of adjunctive procedures such as osseus consolidation and transarterial embolisation. Results Percutaneous thermal ablation has an established role in the successful management of bone lesions, with both curative and palliative intent. This Standards of Practice document provides up-to-date recommendations for the safe performance of thermal ablation of bone tumours.

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6. Full-Text Sections (Europe PMC Open Access JATS XML)

Introduction

The CIRSE Standards of Practice Committee established a writing group which was tasked with producing up-to-date recommendations for the performance of thermal ablation of bone tumours. It is neither a clinical practice guideline nor a systematic review of the literature. CIRSE Standards of Practice documents are not intended to impose a standard of clinical patient care, but recommend a reasonable approach to, and best practices for, the performance of thermal ablation of bone tumours. Institutions should regularly review their internal procedures for development and improvement, taking into account international guidance, local resources and regular internal morbidity and mortality reviews. A summary of key recommendations on thermal ablation of bone tumours can be found in Appendix 1.

Methods

The writing group, which was established by the CIRSE Standards of Practice Committee, consisted of 5 clinicians with internationally recognised expertise in thermal ablation of bone tumours. The writing group reviewed the existing literature on thermal ablation of bone tumours, performing a pragmatic evidence search using PubMed to search for publications in English and relating to human subjects from 2009 to 2019. Selected studies published in 2020 and 2021 during the course of writing these standards were subsequently included. The final recommendations were formulated through consensus.

Background

Percutaneous thermal ablation is an effective, minimally invasive means of treating a variety of focal benign and malignant osseous lesions [1–5] and should be included in the treatment algorithm alongside surgery, systemic therapies and radiotherapy. The latter have shortcomings, particularly in the management of osseus metastases. Radiation therapy (RT) has been considered the standard treatment for pain relief and local control of bone metastases; however, it is limited by a delayed onset of pain relief, the number of adequately radiosensitive tumour types and an upper dose limit, the latter especially important in residual or recurrent tumours [6]. Post-radiation fracture may occur, the majority in the first 4 months post-treatment, especially after stereotactic beam radiation therapy (SBRT) or stereotactic radiosurgery of lytic lesions [7, 8]. RT may be complicated by osteonecrosis or neural injury [9]. RT may be contraindicated owing to limited cumulative tolerance of surrounding radiosensitive organs, such as bowel or spinal cord. Unlike RT, ablative procedures are effective in reducing pain within 48–72 h of treatment [5, 10] and can be repeated in cases of residual or recurrent disease. Surgical osseous metastasectomy is technically challenging, associated with high

complication and morbidity–mortality rates and requires prolonged recovery, delaying the use of systemic therapies [1]. Surgery is excessively invasive and morbid for small focal lesions such as osteoid osteoma (OO) and for palliating painful osseous metastases.

To determine the role of ablation in individual cases, each patient's overall status and target lesion(s) should be discussed in a multidisciplinary team (MDT) setting involving the appropriate subspecialty physicians, specifically to assess the suitability and feasibility of a thermal ablative approach, to select the most appropriate technique and to set the goals of treatment, i.e. curative in benign and certain limited malignant settings (including oligometastatic, oligorecurrent or oligoprogressive disease) and palliative in the remaining malignant settings (including the prevention of skeletal-related events [11]). This discussion is especially important in oligometastatic disease and in spinal lesions where multiple treatment modalities may be required to achieve pain relief and stabilisation.

This document will presume the indication for treatment is clear and approved by the MDT and will define the standards required for the performance of each modality. The advantages of each modality in particular settings will be discussed. Standards for the use of tissue monitoring and protection in preventing non-target destruction of adjacent sensitive structures will be emphasised, as will the optimal timings with respect to other interventions, e.g. embolisation in hypervascular lesions or osseous consolidation [2, 3].

Treatment with Curative Intent:

For benign tumours: Successfully treated lesions include OO, osteoblastoma, eosinophilic granuloma, chondroblastoma and aneurysmal bone cyst [12–20].

For primary malignant bone tumours in carefully selected cases of small (< 3 cm), slow-growing lesions in non-surgical candidates or in patients refusing surgery.

In selected patients presenting with oligometastatic, oligorecurrent and oligoprogressive disease, i.e. < 3–5 potentially treatable metastases, each < 3 cm [21], including ablation specifically directed at preventing compromise of adjacent critical structures due to tumour progression, particularly in spinal lesions [22].

Ablation is offered with palliative intent to treat painful metastases refractory to, or unsuitable for, pharmacologic management, RT or surgery [3, 6, 23–25].

Patient Preparation

Pre-procedural imaging is directed at choosing the optimal approach to the lesion and identification of at-risk structures to plan the deployment of the probe and tissue monitoring/protection devices. Planning studies should be no more than one month old; re-imaging closer to the procedure may be necessary in lesions with aggressive tumour biology or if a patient's symptoms have changed since the most recent imaging.

For osseus detail, non-contrast, thin section (1–3 mm) computed tomography (CT) with reconstructions in multiple planes is the recommended pre-treatment imaging for all osseus lesions [35–37]. Specifically with respect to OO, MRI is limited by lower sensitivity compared to CT in the clear demonstration of the nidus, as the latter can be isointense to cortical bone. Higher sensitivity and specificity rates are achieved with MRI using dynamic contrast enhancement and improved spatial resolution techniques [36, 38, 39]. Technetium-based radionuclide studies will show intense nidus uptake [36]. Contrast-enhanced CT and MRI followed by percutaneous biopsy may be necessary for the pre-treatment evaluation of other benign osseous tumours.

Multiparametric MRI (including T1- and T2-weighted sequences with and without fat saturation, diffusion-weighted imaging and dynamic contrast-enhanced multiplanar T1-weighted sequences) is recommended for the evaluation of malignant lesions [40]. Apparent diffusion coefficient (ADC) and dynamic contrast-enhanced sequences provide useful functional information [37].

Positron emission tomography-CT (PET-CT) will confirm or refute disease as oligometastatic and demonstrates prognostic metabolic parameters, e.g. standardised uptake value and metabolic tumour volume [41]. When PET-CT is performed, bone scans may be considered optional, although technetium-based studies remain of higher sensitivity for osseus metastases [40, 41].

Treatment planning for MR-guided HIFU (MRgHIFU) involves a number of steps: calibration, loading, segmentation, planning and verification, the details of which are beyond the scope of this document; please see the excellent descriptions by Napoli [26].

All the heat-based modalities are painful, and although some ablations may be performed under conscious sedo-analgesia [26, 35, 42], anaesthetic support is invaluable, including general anaesthesia (GA), for the treatment of exquisitely painful targets such as the nidus of OO, and in the treatment of small target lesions where patient movement could jeopardise the ablation or increase the risk to adjacent at-risk structures [35]. The presence of an anaesthetist increases technical and clinical efficacy and reduces complication rates. An immobile target is especially crucial for the remote targeting required for MRgHIFU; thus, in the treatment of OO in children, general anaesthetic is indicated. Similarly, general anaesthesia is indicated for MRgHIFU of upper trunk lesions [26] and for all ablations in children younger than 14 [35].

Pre-procedural anaesthetic review is recommended to assess the patient's performance status and suitability for general anaesthetic, and to evaluate the feasibility of other techniques such as spinal anaesthesia (spinal, pelvic or proximal lower limb lesions) or regional nerve blocks in the peripheries [3]. The latter blocks may also be used to achieve prolonged post-procedural pain relief.

The final choice will depend on patient factors such as age, cardiovascular and respiratory status, renal function, and on lesion factors such as location, size and expected duration of procedure [26, 42].

Treatment

Imaging is used for immediate pre-procedure planning, targeting and intra-procedural guidance. Fluoroscopy with the addition of cone-beam CT technology, CT (with or without CT-fluoroscopy) or MRI can be used for guidance either alone or in combination with 3D navigation or image-fusion systems. Where available, high-resolution multimodality image guidance with CT and fluoroscopy is particularly useful (e.g. in the treatment of spinal lesions) allowing quick and precise probe placement and the deployment of protective/monitoring devices, resulting in lower complication rates and facilitating the combination of additional procedures such as sequential application of embolisation, ablation and percutaneous screw fixation or cementoplasty when indicated [2, 3, 25].

A small proportion of lytic tumours may be targeted under US guidance alone if they are relatively superficial and where adjacent sensitive structures can be clearly seen and avoided. Specifically for HIFU, MR guidance (MRgHIFU) is preferred to ultrasound guidance in the treatment of bone lesions for its precise targeting, and when used, an immediate pre-treatment MRI should be performed in the expected treatment position, to refine the treatment approach [26].

Ideally patients should be provided with patient-controlled analgesia (PCA), particularly if being treated under conscious sedation. For long procedures and lesions in challenging locations, a urinary catheter is recommended. The patient is positioned to optimise the approach to the lesion, achieving the shortest straightest possible path while avoiding non-target sensitive structures. Antibiotic prophylaxis is recommended when cement injection and/or screw placement is planned in combination with the ablation; otherwise, prophylaxis is not routinely indicated but may be used depending on operator and institutional protocols. Vital signs should be monitored throughout.

The CIRSE checklist should be completed [43]. After local asepsis and anaesthesia to skin, subcutaneous tissues and periosteum (including under GA, so-called pre-emptive anaesthesia, shown to reduce post-operative pain [44]), an 11- or 13-gauge bone trocar is inserted to the lesion of interest under image guidance and satisfactory position confirmed. The ablation probe is introduced through the coaxial trocar which is then withdrawn to a point outside the expected ablation zone (~ 1 cm) to prevent contact between the active tip and the trocar [5, 32] so as to avoid a skin/subcutaneous fat burn arising from heat transmission via the trocar. Once satisfied with the relative positions of the probe and trocar, the ablation may proceed. At completion, 5–10 ml of ropivacaine 2 mg/ml are injected to the periosteum to reduce post-procedural pain [32]. Imaging is repeated at completion to evaluate for immediate complications.

In the spine, small tumours located in the ipsilateral half of the vertebral body may be entirely ablated with a unipedicular approach, whereas for tumours extending beyond the sagittal midline of the vertebral body, ablation with a bipedicular approach is recommended to achieve adequate margins.

RFA probes are metallic needles with exposed cutting (active) tips of varying and adjustable length with an insulated shaft. Straight probes are used for small discrete targets such as the nidus of an OO and expandable/multi-clustered (umbrella-configuration) devices are used in the treatment of lesions where a larger ablation volume is required. RFA probes may be monopolar (requiring grounding pads) or bipolar (grounding pads unnecessary).

Specifically for spinal metastases, bipolar radiofrequency electrodes (articulating or straight) are available with thermocouples (incorporated or separate) that can easily combine tissue protection, ablation and vertebral augmentation in a single treatment.

MWA is performed with a water-cooled interstitial antenna. No grounding pads are required and there is no interference with implanted electrical devices such as pacemakers. MWA energy profiles are based on the manufacturer's recommended combination of power level (i.e. wattage) and exposure time (i.e. minutes) needed to obtain the desired size of the ablation area. For larger lesions, MWA can be performed with multiple applicators [33] to achieve overlapping ablation zones. For superficial OO, it is not necessary to penetrate the cortical bone as, if the applicator tip is positioned directly over the lesion, the intense heat will penetrate the cortex and effect ablation. In cases of cortical interruption over or adjacent to the lesion, the probe can be positioned directly into the lesion; however, in most cases, and particularly in osteoblastic lesions, probe delivery is coaxial via a large-bore trocar (11–13G) [32].

Medication and Peri-Procedural Care

Post-procedural pain should be anticipated and is typically maximal in the first 24 h; however, the incidence is significantly higher and the intensity of pain experienced greater following heat-based treatments. Systemic analgesic medications, e.g. parenteral paracetamol, should be prescribed as a routine in addition to anti-inflammatories and should be administered prior, during and post-intervention. Following MRgHIFU ablation of OO in children, 3 doses of betamethasone (4 mg every 12 h) are administered to reduce post-ablation inflammation [35]. Patient-controlled analgesia (PCA) containing an infusion of opioid analgesia and anti-emetic medication is beneficial post-procedure. Post-procedural pain is less commonly reported following cryoablation due to its relative anaesthetic effect.

Patients are usually kept in hospital overnight to ensure adequate recovery from anaesthesia and optimal pain management, especially true for children [26]. The majority of patients will be discharged on the first post-operative day. Even when combining ablation with augmentation techniques in weight-bearing locations, early mobilisation is encouraged; however, advice should be given to avoid excessive stressful weight-bearing and strenuous activity for 2–3 weeks.

Post-Procedural and Follow-Up Care

Early follow-up consists of clinical evaluation immediately, 4 h, 1 day and 1 week post-ablation in order to exclude early complications and to assess the adequacy of the treatment with respect to pain relief.

Delayed follow-up is performed regularly in the months to years following ablation and consists of a thorough interventional radiology outpatient clinic (IROC) assessment as well as biological testing, including tumour markers to ensure clinico-biological stability [59, 60].

Imaging is targeted at (a) the ablated lesion to assess response and exclude residual or recurrent disease, or complications such as infection and (b) staging of the underlying pathology, including the development of progressive disease elsewhere or metachronous oligometastases.

In the curative setting, MRI and PET-CT are the preferred techniques to detect local residual tumour or early relapse [4, 38, 39, 61]. In the palliative setting, clinical follow-up suffices, unless complications are suspected. Interval imaging may be triggered at any point if a complication or recurrence is suspected. Suspicion of

fracture should be evaluated with radiographs and CT. Osseus remineralisation on CT is considered an indicator of successful ablation [26]. MRI and CT are typically performed at 1, 3, 6 and 12 months following MRgHIFU.

Paediatric patients who have undergone MRgHIFU for OO are followed with unenhanced MRI at 6, 12 and 24 months looking for changes in marrow oedema and inflammation in the soft tissues adjacent to the lesion [35].

When performed for pain relief, post-procedural pain should be assessed using a validated tool such as a visual analogue score. Analgesic requirements should be documented and the brief pain inventory quality-of-life score measured at 1, 3, 7 and 14 days post-treatment, and at 30-day intervals for one year following treatment [26]. In children, the response should be assessed using the Faces Pain Score—Revised, analgesic requirements ± the Paediatric Initiative on Methods, Management and Pain Assessment in Clinical Trials (PEDIMMPACT) [35, 62, 63].

Clinically significant complications are not common (approx. 2.3%) following RFA of bone lesions, fracture being the most common (1.8%). Other rarely encountered complications include infection, skin burn, peripheral sensory or motor neuropathy, arthropathy and haematoma [51]. Factors associated with a higher complication rate are a tumour size > 3 cm and previous radiotherapy, the latter the only risk factor specifically associated with an increased minor complication rate [51]. Although rare, neural thermal injury is one of the more frequent complications following RFA. The majority of patients will recover completely from such unintended ablation; however, they must be appropriately advised regarding the likely duration of recovery (between 6 and 18 months).

Both RFA and MWA can cause iatrogenic skin burns [5, 32] if a non-insulated applicator or trocar is used. A systematic review of MWA revealed clinically significant complications, including skin burns in 10/249 (4.0%) patients [64].

Outcomes

Thermal ablation of appropriately selected osseus lesions of varied histologies is associated with high success and low recurrence and complication rates, summarised in Table 1.

Outcomes following thermal ablation of osseus lesions

Success rates are uniformly high for all modalities treating OO. Experience with other benign bone tumours, e.g. ABC, osteoblastoma, is limited [15–20, 83].

Good tumour-free survival rates (67–68% at one year) are reported for RFA, MWA and CA in oligometastatic metastases, including sarcoma [6, 21, 76]. Thermal ablation achieves effective pain relief [4, 5, 25, 48, 64, 66, 77–82, 84]. Multiple studies have found improved pain relief in spinal lesions when ablation was combined with vertebroplasty and/or RT [85–87]. In general, tumours less than 2 cm with no cortical erosion fare better [76]. The combination of cryoablation and bisphonates has been found to be synergistic in the setting of painful osseus metastases [88].